

Final Project: Streaming Demand Signal Model for Content Investment Decisions

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Introduction

This paper presents a data-driven approach to improving content investment decisions for streaming platforms. Streaming platforms such as Netflix, Amazon Prime, and Disney+ must decide what content to invest in long before it reaches audiences. These decisions are often based on past successes, intuition, or limited market signals. While this approach can work in stable markets, it becomes less effective when audience preferences shift quickly. As a result, platforms may fail to recognize emerging trends early enough, leading to missed opportunities.

The purpose of this project is to build a decision-support system that identifies narrative themes showing strong audience interest and connects those themes to specific content opportunities. The central question guiding this project is: which narrative themes are currently gaining audience attention, and which specific titles should streaming platforms consider adapting? To answer this question, the project integrates multiple data sources, including Google Trends to measure audience demand, IMDb datasets to approximate production activity, and title-level data to provide concrete recommendations. The goal of this project is not to predict success, but rather to help decision-makers identify where opportunities may exist before competitors fully respond.

Business Background

Unlike theatrical releases, which depend heavily on opening weekend performance, streaming platforms rely on sustained engagement over time. Content must continue attracting viewers weeks or even months after release. Because of this, timing plays a critical role in determining the success of a new show or movie. A key challenge in this process is the delay between when audience interest begins to increase and when production decisions are made. Developing content takes time, and by the time a show is released, the trend may already be saturated. This delay creates a gap between demand and supply, which can result in lost revenue opportunities.

To better understand how content performs across different platforms, an analysis comparing streaming and theatrical releases was conducted. The findings show that while theatrical releases tend to generate higher short term revenue spikes, streaming content provides more consistent engagement over time. This reinforces the importance of identifying trends early, as long term engagement is a major driver of value for streaming platforms.

Figure 1. *ROI Distribution Comparison Between Streaming and Theatrical Releases*

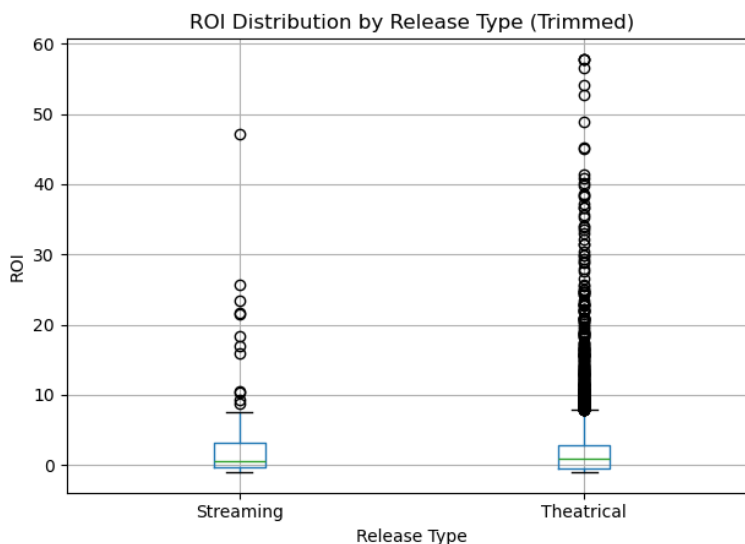


Figure 1 compares streaming and theatrical performance, showing that theatrical releases generate higher peak revenue while streaming content provides more consistent engagement over time. This reinforces the importance of identifying demand trends early, as streaming platforms are able to capture value over a longer period when they respond to emerging audience interests. Because streaming platforms rely more on long-term audience engagement rather than short term performance, timing becomes especially important. If platforms are slow to recognize emerging trends, they risk entering the market after audience interest has already peaked.

Project Objective

The primary objective of this project is to develop a system that translates raw data into actionable insights for content investment. Specifically, the model seeks to identify which narrative themes are gaining audience attention, measure how much content already exists in those areas, and determine whether a gap exists between demand and supply.

In addition to ranking themes, the model also generates title-level recommendations. This step is important because it connects high level insights to specific content ideas. Instead of simply identifying a popular genre, the model provides a list of books that could potentially be adapted into shows or movies. This allows decision-makers to move from analysis to action in a more structured and efficient way.

Data Sources

This project uses three primary data sources, each contributing a different perspective to the analysis. The first data source is Google Trends, which is used to measure audience demand. The data was collected by manually downloading a CSV file from the Google Trends website. The keywords used include “fantasy books,” “science fiction books,” “dystopian books,” “historical fiction books,” and “horror books.” These keywords were selected to represent a

range of narrative themes with different levels of popularity and audience engagement. The dataset provides weekly search interest values on a scale from 0 to 100, allowing the model to track how demand changes over time.

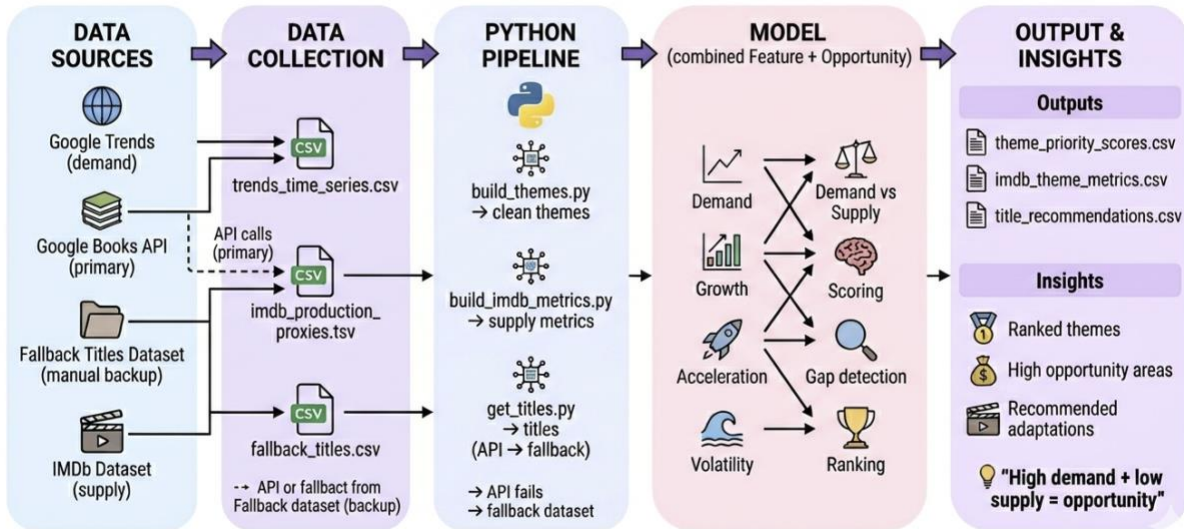
The second data source is the IMDb dataset, which is used as a proxy for content supply. The dataset was obtained from the official IMDb data repository and includes files such as `title.basics.tsv.gz` and `title.ratings.tsv.gz`. These files contain information about movies and television shows, including their genres and ratings. By filtering this data based on relevant keywords, the model estimates how much content has been produced within each theme. This provides a way to approximate supply in the market.

The third data source consists of book titles used for generating recommendations. Two methods were implemented to retrieve this data. When possible, the Google Books API is used to automatically collect titles related to the selected theme. This allows the model to dynamically generate recommendations based on current data. However, the API is subject to rate limits, which can cause requests to fail after multiple attempts. To address this limitation, a fallback dataset was created using publicly available sources such as Google Books search results, Goodreads lists, and Amazon bestseller rankings. Titles were selected based on their relevance to the theme, popularity among readers, and indicators of adaptation potential. This ensures that the pipeline remains functional even when the API is unavailable, while also maintaining transparency in how the dataset was constructed.

Data Pipeline

Figure 2. End-to-End Data Pipeline for Streaming Demand Signal Model

Streaming Demand Signal Model for Content Investment Decisions



The pipeline begins with data collection, where datasets from Google Trends, IMDb, and book title sources are gathered to represent different aspects of the content ecosystem. Google Trends data is manually downloaded as a CSV file and serves as the primary source of audience demand signals. The IMDb dataset, which is downloaded in compressed TSV format, is used to approximate content supply by capturing information about existing movies and television shows. For title-level data, the system attempts to retrieve relevant book titles dynamically using the Google Books API. However, because the API is subject to rate limits, a manually curated fallback dataset is used when API requests fail, ensuring that the pipeline remains stable and reproducible.

Once the data is collected, it is processed through a series of Python scripts that transform raw inputs into structured datasets. The Google Trends data is cleaned to remove inconsistencies and reorganized into a time-series format that allows for trend analysis. From this dataset,

multiple features are derived, including average demand, recent demand, growth (slope), acceleration, and volatility. These features capture not only the level of interest in a theme but also how that interest is changing over time.

At the same time, the IMDb dataset is filtered based on relevant keywords to isolate content associated with each narrative theme. The filtered data is then aggregated to estimate production activity, which serves as a proxy for supply. This step is important because it allows the model to compare audience demand with the amount of existing content in the market.

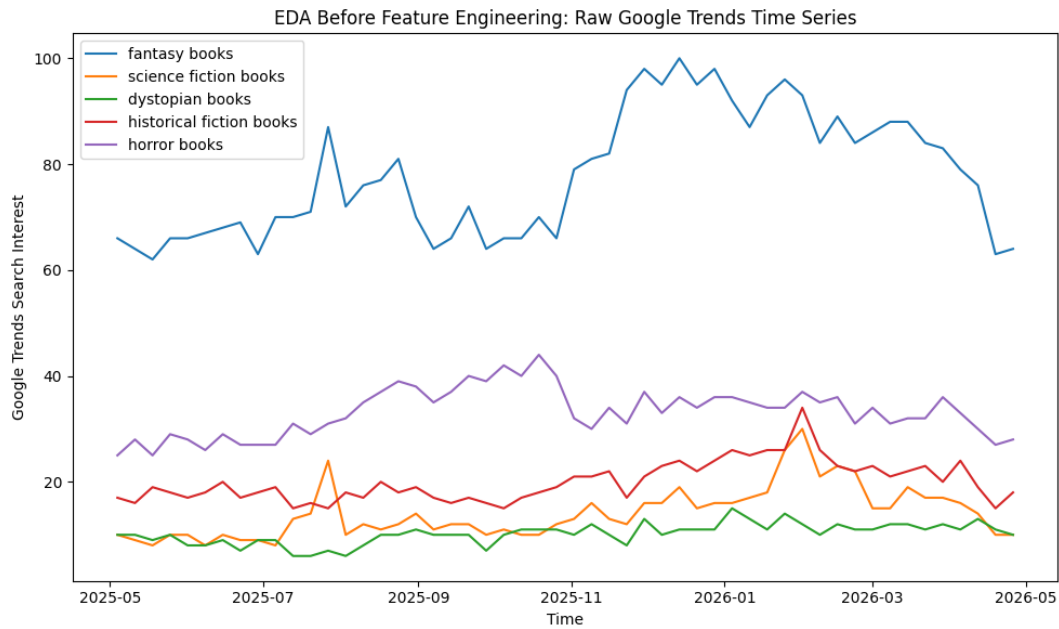
After preprocessing, the transformed datasets are combined and passed into the model. The model normalizes the feature values and applies a weighted scoring approach to generate a priority score for each theme. This scoring process emphasizes themes that exhibit strong demand signals and positive growth patterns. The model then performs a supply–demand gap analysis by comparing the demand-based priority score with the estimated production activity from IMDb. Themes where demand significantly exceeds supply are identified as high-opportunity areas.

The final stage of the pipeline generates output files, including theme priority scores, supply metrics, and title-level recommendations. These outputs provide both high-level insights and actionable recommendations for decision-makers. Additionally, intermediate CSV files are saved at each stage of the pipeline, which improves transparency and ensures that the entire process can be reproduced and validated step by step.

Exploratory Data Analysis (EDA)

Exploratory data analysis was conducted to better understand the structure and patterns within the dataset before building the model.

Figure 3. *Raw Google Trends Time Series for Narrative Themes (Before Feature Engineering)*



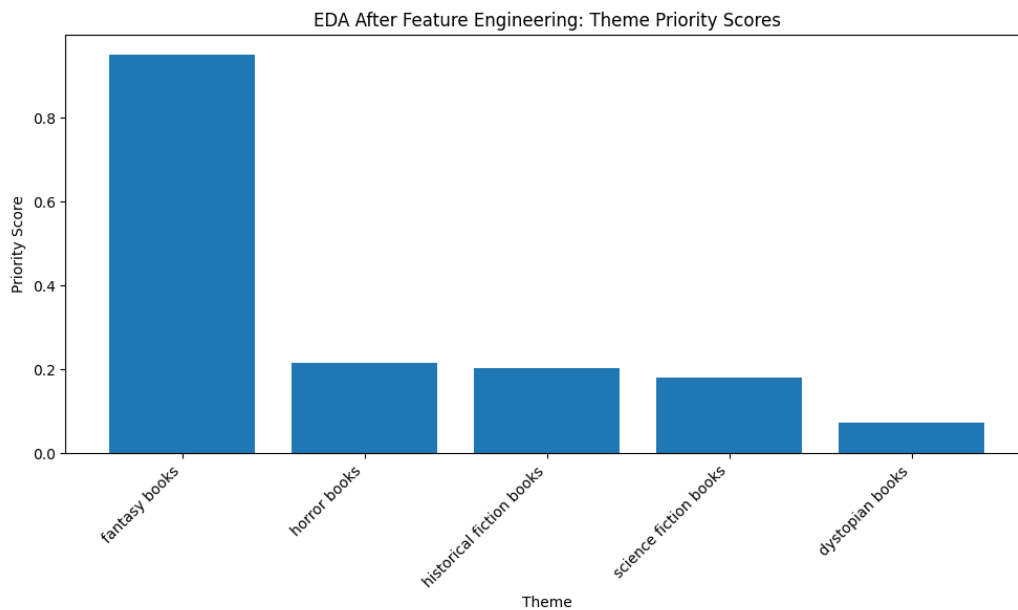
The raw time-series data reveals clear differences in audience demand across the themes analyzed. As shown in Figure 3, fantasy books consistently exhibit the highest levels of search interest over time, indicating strong and sustained audience engagement. In contrast, dystopian books remain relatively low throughout the period, suggesting weaker overall demand. Other themes, such as science fiction, historical fiction, and horror, fall between these extremes, with moderate but more variable levels of interest.

Beyond overall demand levels, the time-series patterns also highlight important differences in trend behavior. Fantasy books not only maintain high demand but also demonstrate periods of upward momentum, particularly in the latter half of the timeline, indicating increasing interest. Science fiction and historical fiction show more fluctuation, with occasional spikes that may reflect temporary events or releases rather than sustained growth. Horror exhibits moderate demand with relatively stable patterns, while dystopian themes show minimal growth or acceleration over time. These differences suggest that demand is not only about how popular a

theme is at a given moment, but also how that popularity evolves. Themes with strong upward trends or increasing momentum may represent emerging opportunities, even if their current demand levels are not the highest. Conversely, themes with flat or declining trends may indicate saturated or declining interest.

To capture these dynamics more effectively, additional features were engineered from the raw time-series data. These include average demand, recent demand, growth (slope), acceleration, and volatility. By incorporating both the level and direction of demand, these features allow the model to distinguish between stable popularity and rising trends, providing a more nuanced understanding of audience behavior. These engineered features serve as key inputs to the scoring model, enabling more accurate identification of high opportunity themes.

Figure 4. *Feature-Engineered Theme Priority Scores Based on Demand Signals*



These features include average demand, recent demand, growth (measured as the slope of the trend), acceleration (the rate of change in growth), and volatility (consistency of demand). Together, these features allow the model to capture not only how popular a theme is at a given

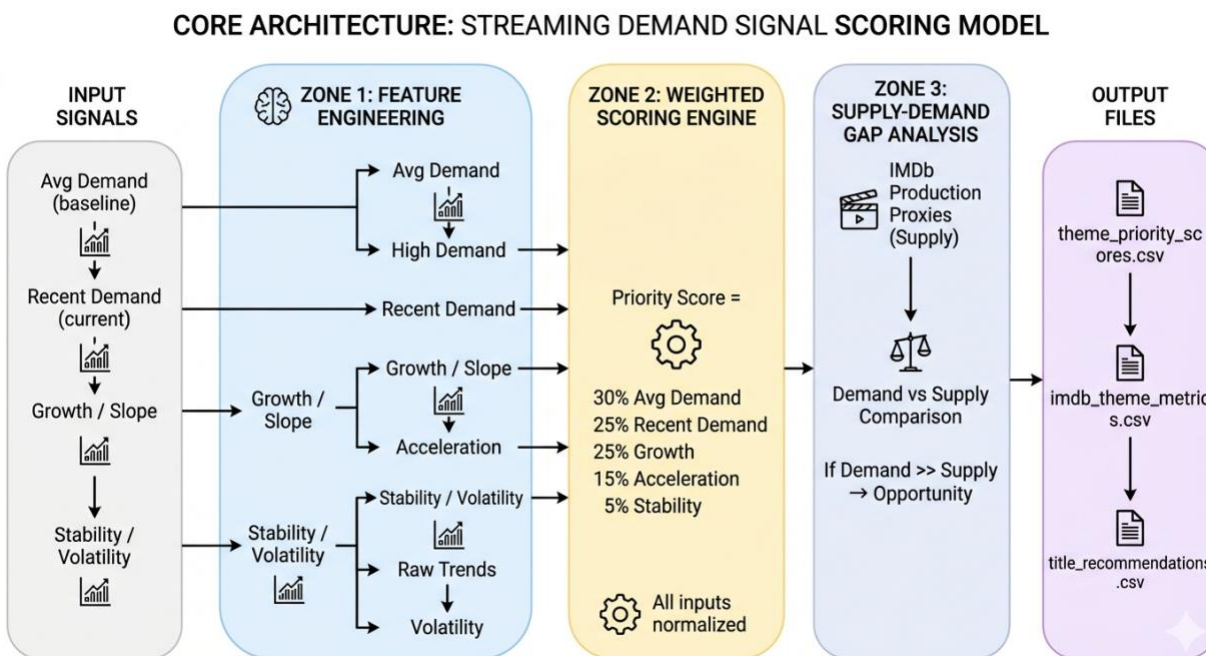
point in time, but also how its popularity is evolving over time. This is important because a theme with moderate demand, but strong growth may represent a better opportunity than a theme with high but declining demand. As shown in Figure 4, the feature engineered priority scores highlight clear differences in opportunity across themes. Fantasy books receive a significantly higher priority score compared to all other categories, reflecting their strong demand levels, consistent growth, and relatively stable interest over time. In contrast, dystopian books receive the lowest score, which aligns with their weaker demand signals and lack of meaningful growth observed in the raw time-series data.

Other themes, such as horror, historical fiction, and science fiction, fall in the middle range, indicating moderate levels of opportunity. While these themes may exhibit some positive signals, such as occasional increases in demand or short-term spikes, they lack the sustained momentum and consistency seen in higher-ranking themes. This demonstrates how the model differentiates between temporary fluctuations and more meaningful, sustained trends.

Importantly, these results show that the model is not solely driven by average demand. Instead, it integrates multiple dimensions of audience behavior to produce a more balanced evaluation. For example, a theme with stable but non-growing demand may receive a lower score than a theme that is actively gaining popularity. This ensures that the model prioritizes forward-looking opportunities rather than relying only on historical popularity. Overall, the feature engineering step transforms raw time-series data into structured signals that can be quantitatively evaluated, allowing the model to identify themes that are not only popular, but also growing and underrepresented. These priority scores serve as the foundation for the subsequent supply demand gap analysis and final recommendation process.

Model Architecture

Figure 5. *Model Architecture: Multi-Stage Demand Scoring and Supply–Demand Gap Analysis*



In the first stage, raw demand data is transformed into meaningful features that capture different dimensions of audience behavior. Average demand represents the overall level of interest in a theme over time, providing a baseline measure of popularity. Recent demand reflects more current audience attention, allowing the model to emphasize emerging trends. Growth, measured as the slope of the time series, indicates whether interest in a theme is increasing or decreasing, while acceleration captures how quickly that change is occurring. Volatility measures the consistency of demand, helping distinguish between stable trends and short term fluctuations. Together, these features provide a comprehensive view of both the level and direction of audience interest.

In the second stage, these features are normalized and combined using a weighted scoring approach to generate a single priority score for each theme. Normalization ensures that all features are comparable despite differences in scale, preventing any single metric from

disproportionately influencing the results. The weighting scheme reflects the relative importance of each feature in identifying opportunity. Higher weights are assigned to features such as average demand and growth, which signal strong and increasing interest, while lower weights are assigned to features like volatility, which primarily capture stability. This approach allows the model to prioritize themes that demonstrate not only high demand, but also positive momentum and sustained engagement over time. Importantly, the weighted scoring model is designed to balance multiple signals rather than rely on any single metric. A theme with high demand but declining growth may receive a lower score than a theme with moderate demand but strong upward momentum. Similarly, themes with highly volatile patterns may be deprioritized compared to those with more stable and consistent growth. This ensures that the model focuses on forward-looking opportunities rather than simply reflecting historical popularity.

In the final stage, the model performs a supply demand gap analysis by comparing the demand-based priority score with production activity derived from IMDb data. The IMDb dataset serves as a proxy for content supply, capturing the number and type of existing productions associated with each theme. By comparing demand signals with supply levels, the model identifies areas where audience interest is strong but content availability is relatively limited. These gaps represent potential opportunities for new content investment.

Themes where demand significantly exceeds supply are classified as high-opportunity areas. This step is critical because it moves beyond measuring popularity and instead evaluates market imbalance, which is a more direct indicator of potential value. In this way, the model shifts the focus from “what is popular” to “where opportunities exist,” aligning the analysis more closely with real-world decision-making needs.

Overall, this architecture integrates feature engineering, weighted scoring, and gap analysis into a structured decision-support framework. By combining multiple dimensions of demand with a supply-side perspective, the model provides a more holistic and actionable approach to identifying high-potential content opportunities.

Model Evaluation

Because this model is designed as a ranking system rather than a predictive model, traditional evaluation metrics such as accuracy, precision, or recall are not applicable. Instead, the model is evaluated based on the consistency, interpretability, and usefulness of its outputs in supporting decision-making.

To assess consistency, the model's results were examined to ensure that themes with stronger demand signals, such as higher average demand and positive growth trends; consistently received higher priority scores. The results confirm this behavior, as themes like fantasy books, which exhibit both strong demand and upward momentum, are ranked significantly higher than themes with weaker or declining trends, such as dystopian books. This indicates that the model is functioning as intended and appropriately weighting key demand signals.

In addition, the model was evaluated for its ability to distinguish between different types of demand patterns. Themes with short term spikes but low overall growth were assigned lower scores than those with sustained upward trends, demonstrating that the model prioritizes long term momentum over temporary fluctuations. This behavior is important for identifying meaningful opportunities rather than reacting to short lived interest.

To further validate the model, its outputs were compared with observable industry trends. The high ranking of fantasy content aligns with the increasing number of fantasy adaptations

across major streaming platforms in recent years. This consistency suggests that the model is effectively capturing real-world demand patterns and translating them into actionable insights.

Overall, these evaluation steps demonstrate that the model produces logical, consistent, and interpretable rankings, making it a useful tool for identifying high opportunity themes in a content investment context.

Results

The results of the model highlight clear differences across the themes analyzed, demonstrating how variations in demand signals translate into different levels of opportunity. Among the themes evaluated, fantasy books received the highest priority score, reflecting a combination of strong overall demand, sustained recent interest, and positive growth over time. This indicates that audience attention toward this theme is not only high, but also actively increasing, making it a strong candidate for emerging opportunity.

In contrast, themes such as dystopian books and historical fiction received lower priority scores due to weaker demand signals, including lower average search interest and slower or less consistent growth patterns. While these themes may still have established audiences, the data suggests that they are not currently experiencing the same level of momentum as higher-ranked categories.

A key insight from the model is that high demand does not always correspond to high production activity. By incorporating IMDb data as a proxy for content supply, the model identifies situations where audience interest is strong but existing content remains relatively limited. This imbalance between demand and supply is critical, as it highlights areas where the market may be underserved.

Themes with high demand but lower production representation represent the most promising opportunities for content investment. From a strategic perspective, these are the areas where early action can provide a competitive advantage, allowing platforms to capture audience interest before the market becomes saturated.

Overall, the results demonstrate that the model effectively distinguishes between stable, saturated themes and those that are actively gaining momentum. By combining demand trends with supply conditions, the model provides a more forward looking and actionable view of opportunity, enabling decision-makers to prioritize investments based not only on current popularity, but also on future growth potential.

Application of Model

In practice, this model can be used as a decision-support tool within content strategy teams at streaming platforms. The process begins by running the model on updated demand data, which produces a ranked list of themes based on their priority scores. This allows decision-makers to quickly identify high potential areas and focus their attention on themes that demonstrate strong demand signals and growth momentum.

Once a theme is selected, the model generates a set of recommended titles associated with that theme. These titles serve as starting points for further evaluation, including feasibility analysis, licensing availability, and production planning. This step bridges the gap between high-level trend identification and actionable content decisions, enabling teams to move from abstract insights to specific investment opportunities.

The model can also be applied on a recurring basis. By updating the input data regularly, companies can monitor how demand signals evolve over time and adjust their strategies

accordingly. This enables a more proactive approach to content planning, where decisions are based on emerging trends rather than reactive responses to already saturated markets.

From an operational perspective, the model supports prioritization across a portfolio of potential investments. Instead of relying on intuition or isolated signals, decision-makers can use a structured, data-driven framework to compare opportunities across multiple themes and allocate resources more effectively.

In a real-world setting, this approach can be integrated into a broader analytics ecosystem, where it complements additional data sources such as user viewing behavior, platform engagement metrics, and competitive intelligence. This integration allows organizations to combine external demand signals with internal performance data, creating a more comprehensive foundation for strategic decision-making.

Title-Level Recommendations

After identifying the highest priority theme, the model generates title-level recommendations to support content development decisions. This step translates high level insights into specific, actionable opportunities by linking themes to individual book titles.

Figure 6. *Title-Level Recommendations for High-Priority Theme (Fantasy Books)*

title	author	source	notes
The Name of the Wind	Patrick Rothfuss	Goodreads	Highly requested
Fourth Wing	Rebecca Yarros	Goodreads	High popularity
Iron Flame	Rebecca Yarros	Amazon	Sequel trending
The Priory of the Orange Tree	Samantha Shannon	Goodreads	High rating
A Court of Thorns and Roses	Sarah J. Maas	Goodreads	Strong fanbase
Dune	Frank Herbert	Classic	Already adapted
Shadow and Bone	Leigh Bardugo	Netflix tie-in	Proven adaptation
The Atlas Six	Olivie Blake	Goodreads	Recent viral
The Poppy War	R.F. Kuang	Amazon	Dark fantasy rising
Mistborn	Brandon Sanderson	Goodreads	Popular series

As shown in Figure 6, the model produces a ranked list of titles based on their relevance, popularity, and alignment with the selected theme. Examples include well known works such as *The Name of the Wind*, *Fourth Wing*, and *Iron Flame*, which exhibit strong audience interest and engagement signals. These recommendations serve as starting points for further evaluation, including feasibility, licensing, and production considerations. Rather than making final decisions, the model helps narrow the search space and directs attention toward high-potential content candidates.

Financial Interpretation

From a financial perspective, the value of this model is grounded in observed performance patterns from prior analysis of streaming and theatrical content. That analysis showed that content associated with higher audience interest and engagement levels tends to generate stronger performance outcomes, including higher revenue distributions and sustained viewer activity. While theatrical releases often produce short-term revenue spikes, streaming content derives value from long-term engagement, subscriber retention, and repeat viewership. By linking these findings to the current model, themes identified with strong demand signals, such as high search interest, positive growth trends, and increasing momentum; can be interpreted as having a higher likelihood of generating favorable performance outcomes. In contrast, themes with weaker or declining demand are less likely to deliver the same level of engagement or financial return.

The model does not attempt to predict exact revenue figures. Instead, it identifies conditions that have historically been associated with strong financial performance. In particular, themes that exhibit high demand combined with relatively low production activity represent situations where audience interest has not yet been fully captured by the market.

These demand supply gaps are especially important from an investment perspective. They indicate areas where early entry may provide a competitive advantage, allowing platforms to capture audience interest before the market becomes saturated. As a result, these opportunities are more likely to generate above-average returns compared to investments in already crowded or declining categories. Overall, this approach provides a defensible link between data signals and financial value by grounding the model's outputs in observed industry behavior rather than unsupported numerical assumptions.

Managerial Implications

This model provides a structured, data driven framework for improving content investment decisions in streaming platforms. Rather than relying primarily on intuition or past success, decision-makers can use demand signals to identify where audience interest is growing and where the market remains under supplied. This allows organizations to make more informed and forward-looking investment choices.

One key implication is improved prioritization of resources. By focusing on themes that exhibit strong demand but limited existing content, companies can allocate investment toward projects with a higher likelihood of generating sustained engagement. This reduces the risk of investing in oversaturated areas while increasing the chances of capturing emerging trends early in their lifecycle.

Another important implication is better timing of decisions. Because the model detects changes in demand before they are reflected in production activity, it enables companies to act earlier in the content development process. Early entry into high growth themes can provide a significant competitive advantage, allowing platforms to establish stronger positioning and capture audience attention before competition intensifies.

In addition, the model supports portfolio-level decision-making. Instead of evaluating content opportunities in isolation, decision-makers can compare multiple themes simultaneously and prioritize investments based on relative opportunity. This leads to more efficient allocation of budgets and resources across a range of potential projects.

Overall, the model supports a shift from reactive decision-making to proactive strategy. By aligning content investments more closely with evolving audience behavior, streaming platforms can improve both the effectiveness and efficiency of their content strategies.

Challenges and Limitations

While the model provides valuable insights, several limitations should be considered when interpreting the results. First, the model relies on proxy data to represent content supply. IMDb datasets are used to approximate production activity, but they may not fully capture all aspects of content availability, particularly for streaming-exclusive releases. As a result, the supply side of the model should be viewed as an estimate rather than a complete representation of the market.

Second, data availability introduces constraints. The Google Books API, used to retrieve title-level data, is subject to rate limits and may fail during execution. To address this, a fallback dataset was manually constructed using publicly available sources such as Google Books search results, Goodreads lists, and Amazon bestseller rankings. While this approach ensures pipeline continuity, it introduces a degree of manual selection that may affect consistency and scalability.

In addition, the model does not incorporate operational or financial constraints such as production cost, licensing availability, or competitive strategy. These factors play a critical role in determining the feasibility and success of content investments and would need to be considered alongside the model's outputs in real-world applications.

Finally, the model is designed to identify relative opportunity rather than predict exact financial outcomes. While this makes the approach more robust and aligned with real-world uncertainty, it also means that the results should be interpreted as directional guidance rather than definitive recommendations. These limitations highlight opportunities for future improvement, particularly through the integration of richer data sources and more comprehensive modeling approaches.

Future Directions

This model can be further improved by incorporating additional data sources that more directly reflect audience engagement, such as streaming platform viewership data and social media sentiment. These inputs would provide a more comprehensive view of audience behavior beyond search activity alone.

Future enhancements could also integrate financial and operational constraints, including production budgets, licensing availability, and expected return on investment. Incorporating these factors would allow the model to move beyond opportunity identification toward more complete investment decision support.

In addition, more advanced modeling techniques, such as machine learning-based forecasting, could be explored to improve the accuracy of demand trend estimation and opportunity detection. These extensions would enhance the model's ability to adapt to changing audience dynamics and support more precise strategic planning.

Conclusion

This project demonstrates how data-driven methods can improve content investment decisions in the streaming industry. By combining audience demand signals from Google Trends with production activity proxies from IMDb, the model identifies themes that represent strong opportunities for future development.

A key strength of the approach is its ability to detect gaps between demand and supply. These gaps highlight areas where audience interest is growing but has not yet been fully addressed by existing content, creating opportunities for early and strategic investment.

By extending the analysis to title-level recommendations, the model connects high-level insights to specific content opportunities, enabling decision-makers to move from trend identification to actionable decisions.

Although the model does not predict exact financial outcomes, it provides a structured and defensible framework for prioritizing investments based on observable data patterns. Overall, this approach offers a practical and forward-looking tool for navigating an increasingly competitive content landscape.